

Practical Assignment 1.2

Reasoning in Knowledge Based Expert System

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1. Title

AI Career Advisor Using SWI-Prolog

A rule-based AI chatbot that recommends suitable career options based on a user's interests and skills and provides information about different careers and their required skills.

2. Abstract

I developed an **AI Career Advisor using Prolog**. The system is a simple rule-based expert system that stores career-related knowledge in the form of facts and uses rules to provide recommendations.

The chatbot can recommend careers based on interests such as Artificial Intelligence, Web Development, Data, Cybersecurity and

Networking. It can also recommend careers based on skills such as Python, SQL, Networking and Machine Learning. In addition, it can provide career descriptions and display the skills required for a particular career.

The system uses important Prolog concepts such as **facts, predicates, rules, variables, pattern matching, recursion and logical inference.**

3. Problem Statement

Students often find it difficult to identify suitable career options based on their interests and technical skills.

The objective is to develop a simple AI-based career advisor that:

- accepts a user's interests or skills,
- stores career-related knowledge,
- applies logical rules,
- recommends suitable careers,
- provides career descriptions,
- displays required skills,
- handles unknown inputs, and
- interacts with the user through a chatbot interface.

4.Prolog Code:

```
/*
=====
=
    AI CAREER ADVISOR - IRA

    =====
= */

/*
=====
=
    1. KNOWLEDGE BASE - CAREERS

    =====
== */

career(full_stack_developer).
career(ai_ml_engineer).
career(data_scientist).
career(cybersecurity_analyst).
career(network_engineer).

/*
=====
=
```

2. CAREER DESCRIPTIONS

```
=====
== */
```

```
description(
    full_stack_developer,
    'A Full Stack Developer builds both frontend and backend
    applications.'
).
```

```
description(
    ai_ml_engineer,
    'An AI/ML Engineer develops intelligent systems and machine
    learning models.'
).
```

```
description(
    data_scientist,
    'A Data Scientist analyzes data and extracts useful information from
    it.'
).
```

```
description(
    cybersecurity_analyst,
```

'A Cybersecurity Analyst protects systems and networks from security threats.'

).

description(

network_engineer,

'A Network Engineer designs, manages and troubleshoots computer networks.'

).

/*

=====

=

3. REQUIRED SKILLS FOR EACH CAREER

=====

== */

required_skill(full_stack_developer, html).

required_skill(full_stack_developer, css).

required_skill(full_stack_developer, javascript).

required_skill(full_stack_developer, sql).

required_skill(ai_ml_engineer, python).

required_skill(ai_ml_engineer, machine_learning).

required_skill(ai_ml_engineer, mathematics).

required_skill(data_scientist, python).

required_skill(data_scientist, sql).

required_skill(data_scientist, statistics).

required_skill(cybersecurity_analyst, networking).

required_skill(cybersecurity_analyst, linux).

required_skill(cybersecurity_analyst, security).

required_skill(network_engineer, networking).

required_skill(network_engineer, routing).

required_skill(network_engineer, troubleshooting).

/*

=====

=

4. INTERESTS RELATED TO CAREERS

=====

== */

interest_career(web, full_stack_developer).

interest_career(frontend, full_stack_developer).

interest_career(backend, full_stack_developer).

interest_career(ai, ai_ml_engineer).

interest_career(machine_learning, ai_ml_engineer).

interest_career(robotics, ai_ml_engineer).

interest_career(data, data_scientist).

interest_career(analytics, data_scientist).

interest_career(statistics, data_scientist).

interest_career(hacking, cybersecurity_analyst).

interest_career(security, cybersecurity_analyst).

interest_career(cybersecurity, cybersecurity_analyst).

interest_career(networking, network_engineer).

interest_career(networks, network_engineer).

interest_career(cisco, network_engineer).

/*

=====

=

5. USER SKILL -> CAREER RELATION

=====

== */

skill_career(javascript, full_stack_developer).

skill_career(html, full_stack_developer).

skill_career(css, full_stack_developer).

skill_career(sql, full_stack_developer).

skill_career(python, ai_ml_engineer).

skill_career(machine_learning, ai_ml_engineer).

skill_career(mathematics, ai_ml_engineer).

skill_career(python, data_scientist).

skill_career(sql, data_scientist).

skill_career(statistics, data_scientist).

skill_career(linux, cybersecurity_analyst).

skill_career(networking, cybersecurity_analyst).

skill_career(security, cybersecurity_analyst).

skill_career(networking, network_engineer).

skill_career(routing, network_engineer).

skill_career(troubleshooting, network_engineer).

/*

=====

=

6. CAREER RECOMMENDATION RULE

=====

== */

```
recommend(Interest, Career) :-  
    interest_career(Interest, Career).
```

```
/*
```

```
=====
```

```
=
```

7. RECOMMEND CAREER USING SKILL

```
=====
```

```
== */
```

```
recommend_by_skill(Skill, Career) :-  
    skill_career(Skill, Career).
```

```
/*
```

```
=====
```

```
=
```

8. DISPLAY CAREER DESCRIPTION

```
=====
```

```
== */
```

```
show_description(Career) :-  
    description(Career, Description),  
    write(Description),  
    nl.
```

```

/*
=====
=

9. SHOW REQUIRED SKILLS

=====
== */

```

```

show_skills(Career) :-
    write('Required skills:'),
    nl,
    required_skill(Career, Skill),
    write('- '),
    write(Skill),
    nl,
    fail.

```

```

show_skills(_).

```

```

/*
=====
=

10. SHOW ALL CAREERS FROM A LIST

This is used for multiple recommendations.

=====
== */

```

```
show_careers([]).
```

```
show_careers([Career | Rest]) :-
```

```
    write('- '),
```

```
    write(Career),
```

```
    nl,
```

```
    show_careers(Rest).
```

```
/*
```

```
=====
```

```
=
```

11. CHECK WHETHER A SKILL IS REQUIRED

```
=====
```

```
== */
```

```
skill_required(Career, Skill) :-
```

```
    required_skill(Career, Skill).
```

```
/*
```

```
=====
```

```
=
```

12. START THE CAREER ADVISOR

```
=====
```

```
== */
```

start :-

```
    nl,  
    write('====='),  
nl,  
    write('    AI CAREER ADVISOR USING PROLOG'), nl,  
    write('====='),  
nl,  
    nl,  
  
    write('Welcome to the AI Career Advisor!'), nl,  
    write('I can recommend careers based on your'), nl,  
    write('interests and skills.'), nl,  
    nl,  
  
    write('Available commands:'), nl,  
    write('[recommend,Interest].'), nl,  
    write('[skill_recommend,Skill].'), nl,  
    write('[describe,Career].'), nl,  
    write('[skills,Career].'), nl,  
    write('[help].'), nl,  
    write('[bye].'), nl,  
    nl,
```

advisor_loop.

/*

=====

=

13. MAIN ADVISOR LOOP

=====

== */

advisor_loop :-

write('Advisor:- '),

read(Input),

advisor_reply(Input).

/*

=====

=

14. RECOMMEND CAREER BASED ON INTEREST

Example:

[recommend,ai].

=====

== */

advisor_reply([recommend, Interest]) :-

```

findall(Career, recommend(Interest, Career), Careers),

(
    Careers = []
    ->
    write('Sorry, I do not have enough knowledge about '),
    write(Interest),
    write('.'),
    nl,
    write('Try interests such as:'),
    nl,
    write('web, ai, data, cybersecurity, networking.'),
    nl
    ;
    write('Based on your interest in '),
    write(Interest),
    write(', I recommend:'),
    nl,
    show_careers(Careers)
),

nl,
advisor_loop.

```

```
/*
```

```
=====
```

```
=
```

15. RECOMMEND CAREER BASED ON SKILL

Example:

[skill_recommend,python].

findall/3 collects ALL matching careers.

```
=====
```

```
== */
```

advisor_reply([skill_recommend, Skill]) :-

```
    findall(Career,
            recommend_by_skill(Skill, Career),
            Careers),
```

```
(
```

```
    Careers = []
```

```
->
```

```
    write('I do not have information about the skill: '),
```

```
    write(Skill),
```

```
    nl
```

```
;
```

```
    write('Your skill in '),
```

```
    write(Skill),  
    write(' is useful for:'),  
    nl,  
    show_careers(Careers)  
)
```

```
nl,  
advisor_loop.
```

```
/*  
=====
```

16. DESCRIBE A CAREER

Example:

```
[describe,ai_ml_engineer].
```

```
=====
```

```
== */
```

```
advisor_reply([describe, Career]) :-
```

```
    description(Career, Description),
```

```
    write('Career: '),
```

```
    write(Career),
```

```
    nl,
```

```
    write('Description: '),
```

```
write(Description),  
nl,  
nl,  
advisor_loop.
```

```
/*  
=====
```

=

17. UNKNOWN CAREER

=====

```
== */
```

```
advisor_reply([describe, Career]) :-  
    write('Sorry, I do not know about the career: '),  
    write(Career),  
    nl,  
    nl,  
    advisor_loop.
```

```
/*  
=====
```

=

18. SHOW REQUIRED SKILLS

Example:

```
[skills,ai_ml_engineer].
```

```
=====
== */
```

```
advisor_reply([skills, Career]) :-
```

```
    career(Career),
```

```
    show_skills(Career),
```

```
    nl,
```

```
    advisor_loop.
```

```
/*
```

```
=====
```

```
=
```

```
19. UNKNOWN CAREER FOR SKILLS
```

```
=====
== */
```

```
advisor_reply([skills, Career]) :-
```

```
    write('Sorry, this career is not available in my knowledge base.'),
```

```
    nl,
```

```
    nl,
```

```
    advisor_loop.
```

```

/*
=====
=

20. HELP COMMAND

=====
== */

```

```

advisor_reply([help]) :-

```

```

    nl,
    write('Available commands:'), nl,
    nl,

```

```

    write('[recommend,ai].'), nl,
    write(' -> Recommend a career based on interest.'), nl,
    nl,

```

```

    write('[skill_recommend,python].'), nl,
    write(' -> Recommend careers based on skill.'), nl,
    nl,

```

```

    write('[describe,ai_ml_engineer].'), nl,
    write(' -> Describe a career.'), nl,
    nl,

```

```

    write('[skills,ai_ml_engineer].'), nl,

```

```
write(' -> Show required skills. '), nl,  
nl,
```

```
write('[bye]. '), nl,  
write(' -> Exit the advisor. '), nl,  
nl,
```

```
advisor_loop.
```

```
/*  
=====
```

=

21. EXIT

```
=====
```

== */

```
advisor_reply([bye]) :-
```

```
nl,  
write('Thank you for using AI Career Advisor!'),  
nl,  
write('Good luck with your career!'),  
nl.
```

```
advisor_reply(_) :-
```

```
write('I did not understand that command. '),  
nl,
```

```
write('Type [help]. to see available commands. '),  
  
nl,  
  
nl,  
  
advisor_loop.
```

OUTPUT:

```
SWI-Prolog console  
File Settings Tools Debug GUI Help  
Welcome to SWI-Prolog (threaded, 64 bits, version 10.1.13)  
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.  
Please run ?- license. for legal details.  
  
For online help and background, visit https://www.swi-prolog.org  
For built-in help, use ?- help(Topic). or ?- apropos(Word).  
  
30 ?- ['C:/Users/saksh/OneDrive/Pictures/Documents/prolog/ai_career_advisor.pl'].  
Warning: C:/Users/saksh/OneDrive/Pictures/Documents/prolog/ai_career_advisor.pl:341:  
Warning: Singleton variables: [Career]  
true.  
  
31 ?- start.  
  
=====
```

```
AI CAREER ADVISOR USING PROLOG  
=====
```

```
Welcome to the AI Career Advisor!  
I can recommend careers based on your  
interests and skills.  
  
Available commands:  
[recommend,Interest].  
[skill_recommend,Skill].  
[describe,Career].  
[skills,Career].  
[help].  
[bye].  
  
[recommend,web].  
Based on your interest in web, I recommend:  
- full_stack_developer  
  
|: [recommend,ai].  
Based on your interest in ai, I recommend:  
- ai_ml_engineer  
  
|: [recommend,data].  
Based on your interest in data, I recommend:  
- data_scientist
```

```
|: [skill_recommend,python].
Your skill in python is useful for:
- ai_ml_engineer
- data_scientist

|: [skill_recommend,networking].
Your skill in networking is useful for:
- cybersecurity_analyst
- network_engineer

|: [describe,ai_ml_engineer].
Career: ai_ml_engineer
Description: An AI/ML Engineer develops intelligent systems and machine learning models.

|: [describe,full_stack_developer].
Career: full_stack_developer
Description: A Full Stack Developer builds both frontend and backend applications.

|: [skills,ai_ml_engineer].
Required skills:
- python
- machine_learning
- mathematics

|: [skills,full_stack_developer].
Required skills:
- html
- css
- javascript
- sql

|: [recommend,photography].
Sorry, I do not have enough knowledge about photography.
Try interests such as:
web, ai, data, cybersecurity, networking.

|: [skill_recommend,java].
I do not have information about the skill: java
```

```
|: [describe,pilot].
Sorry, I do not know about the career: pilot

|: [help].|:
Available commands:

[recommend,ai].
-> Recommend a career based on interest.

[skill_recommend,python].
-> Recommend careers based on skill.

[describe,ai_ml_engineer].
-> Describe a career.

[skills,ai_ml_engineer].
-> Show required skills.

[bye].
-> Exit the advisor.

|: [bye].

Thank you for using AI Career Advisor!
Good luck with your career!
true .

3 ?- ▲
```

Conclusion

The **AI Career Advisor using Prolog** successfully demonstrates how Artificial Intelligence can be implemented using a knowledge-based and rule-based approach.

The project helped demonstrate that AI does not always require machine learning. A system can also show intelligent behaviour by using **knowledge representation, logical reasoning and inference**.